Research Proposal on Agricultural Product Quality Optimization and Yield Enhancement Based on Agricultural Sensor Analytics

1. Project Background and Research Objectives

With the advancement of agricultural modernization and digital agriculture, agricultural production is progressively transitioning from a traditional, experience-based management model to a data-driven paradigm. In practical agricultural production, crop yield and quality are heavily influenced by multifaceted factors, including soil environments, meteorological conditions, water and fertilizer management, and the growth process itself. However, most cultivation management still relies on empirical knowledge and lacks continuous, systematic data support. Consequently, it is often difficult to accurately determine which factors truly impact yield and quality, and equally challenging to identify production issues in a timely manner.

In recent years, the development of agricultural sensors, the Internet of Things (IoT), and big data analytics has provided novel solutions for the digital management of agricultural production. By deploying sensor devices in fields, continuous monitoring of soil and environmental dynamics becomes achievable. Through the establishment of agricultural databases and analytical models, key patterns governing crop growth can be uncovered from long-term data accumulation. Furthermore, predictive models and decision support systems can assist farmers in formulating management practices proactively, thereby enhancing production efficiency and economic returns.

This project proposes to construct an agricultural data collection and analytics platform on an experimental field of approximately 10 *mu* (approx. 0.67 hectares). By leveraging agricultural sensors to collect real-time environmental data, combined with manually gathered growth phenotypic data and final yield/quality outcomes, we aim to establish an agricultural big data analytics framework. This system will explore the critical factors driving the formation of agricultural product yield and quality, ultimately developing a scalable and reproducible digital agriculture management model.

2. Agricultural Sensor Deployment Scheme

To comprehensively capture the crop growth environment, this project proposes an agricultural IoT platform comprising a soil monitoring system, a meteorological monitoring system, and a data transmission system.

Soil and Meteorological Monitoring

For soil monitoring, each experimental zone will be equipped with soil moisture, soil temperature, soil electrical conductivity (EC), and soil pH sensors. Specifically:

- The soil moisture sensor monitors changes in soil water content to guide irrigation management.
- The soil temperature sensor monitors the root-zone growth environment.
- The EC sensor reflects variations in soil nutrients and salinity.
- The pH sensor monitors soil acidity and alkalinity levels.

For meteorological monitoring, a compact agricultural weather station will be installed to continuously track indices such as air temperature, air humidity, light intensity, wind speed, wind direction, and rainfall. These indicators reflect the external environmental conditions during crop growth and serve as crucial benchmarks for subsequent analyses of environmental impacts.

Data Transmission and Quality Control

All sensors are connected to a cloud platform via wireless networks. Regarding data transmission and quality control, data collected by each sensor node is aggregated through a 4G/LoRa edge gateway and then transmitted to the Alibaba Cloud IoT platform using the MQTT protocol. The platform utilizes its built-in rules engine to perform real-time parsing of the massive time-series data streams; structured data is stored in MySQL, while high-frequency time-series data streams are asynchronously written into InfluxDB, a distributed time-series database, enabling automated uploads and remote monitoring.

The data collection frequency for this project is set to 30 minutes per interval, rather than 5 minutes or 1 hour. This configuration is a dual optimization based on the energy constraints of Wireless Sensor Networks (WSNs) and the physical inertia of agricultural environmental factors:

1. According to the resampling experiments conducted by Sultana et al., if the sampling interval is extended to 60 minutes, the diurnal curves of soil and canopy temperatures exhibit significant statistical distortion (with the standard deviation bias increasing by 0.33). This causes the decision support system to suffer a warning delay of over 1 hour

when crops encounter high-temperature or salinity stress at midday.

2. Conversely, if high-frequency sampling of 5 or 10 minutes is adopted, the data manifests as flat noise with strong collinearity within short intervals due to the high thermal inertia and physical hysteresis of soil moisture, pH, and EC. This not only fails to contribute additional effective information but also causes the energy consumption of battery-powered field nodes to increase exponentially due to frequent wake-ups of the wireless transmission module, thereby severely shortening the system's life cycle across the entire crop growth duration. The 30-minute interval balances these factors to ensure the environment's dynamic variations are fully captured.

Experimental Field Partitioning and Plot Design

To improve the reliability of subsequent data analysis, the experimental field will undergo partitioned management. This research scheme implements a refined spatial grid and comparative plot engineering design tailored to a 10-*mu* co-cultivation experimental field of rice and saline-alkali land persimmons. The entire field is divided into 12 standard spatial grid units, with a single grid area measuring approximately 0.83 *mu* (23.5 m×23.5 m). The 5-*mu* rice zone and the 5-*mu*persimmon zone each occupy 6 grids. Based on the variogram analysis of co-cultivation field heterogeneity, a square grid of around 0.8 *mu* represents the optimal geometric scale for IoT sensors to capture the spatial evolution laws of soil moisture and nutrients (Han et al., 2025).

In the field comparative experiment design, both crop zones are independently partitioned into:

- **Zone A:** Conventional Experience Control Zone (2 grids).
- **Zone B:** Precision Data-Driven Irrigation Zone (2 grids).
- **Zone C:** Integrated Water-Fertilizer and Salinity Regulation Zone (2 grids).

To evade edge effects, all IoT collection sub-stations are deployed at the geometric center of each grid unit.

Field Operation Procedures

1. Rice Zone (Sensor burial depths: 10 cm and 20 cm): Sensors are buried at depths of 10 cm and 20 cm, where the rice root system is most active.
    - *Water control in Zones B and C:* The system automatically toggles electric valves based on this set of moisture data to achieve automated "shallow, wet, and drying" (AWD) rice irrigation.

- Fertilizer control in Zone C: Upon entering the heading stage, the system works in tandem with a variable-rate fertilization system to precisely flush fertilizer into the field during irrigation (Zhou et al., 2025).

2. Saline-Alkali Land Persimmon Zone (Sensor burial depths: 15 cm and 40 cm): Addressing the tendency of saline-alkali land to undergo soil salinization, sensors are embedded across two layers: the 15 cm topsoil layer and the 40 cm primary root water-absorption layer.

    - Fertilizer and salinity control in Zone C: The setup is connected to an intelligent multi-channel water-and-fertilizer integrated machine. When the deep-layer EC value drops below 2.5 dS/m, indicating a nutrient deficiency, the system automatically injects "sweetness-and-efficiency-enhancing" fertilizer. When the surface-layer EC value exceeds 5.5 dS/m, indicating severe surface salinization, the system immediately cuts off the fertilizer solution and switches to drip irrigation with a large volume of pure water for soil leaching, flushing salt back into deeper layers to prevent root-burning (Martínez-Gomez & Zhao, 2026).

Based on plot locations, soil conditions, and cultivation status, the 10-*mu* experimental field is organized into designated monitoring zones, each equipped with identical sensor configurations and governed by unified data collection standards.

- Automated Subterranean Data (Every 30 minutes): All sensors across the 12 partitions are uniformly configured to upload data automatically every 30 minutes. The cloud platform aligns and packages all field parameters—such as air temperature, soil moisture, and EC values—at identical timestamps, ensuring perfect temporal synchronization for subsequent comparative analysis.

- Manual Above-Ground Growth Observations (Every 3 or 7 days): Routine Periods (Every 7 days): Every Monday morning, technicians inspect the fields to measure rice plant height and node accumulation, as well as count the number of blossoms and leaves on the persimmon trees.
    - Critical Fruiting Period (Every 3 days): During the persimmon fruit expansion stage—the critical period demanding precise water and fertilizer regulation—the observation frequency increases to once every 3 days to measure fruit diameter and fruit count.

    - Daily Mandatory Inspection: Every afternoon at 16:00, technicians walk through the 12 plots specifically to check for pests and diseases, recording a 1 for occurrence and a 0 for non-occurrence.

Through this partitioned management, growth discrepancies across different zones under identical climatic conditions can be compared, enabling the analysis of how soil environmental variations affect crop yield and quality. Concurrently, differentiated management practices are executed in specific zones, such as adjusting irrigation frequencies or fertilization schemes. Taking the water-fertilizer control experiments during the critical fruit expansion stage of alkali persimmons or the grain-filling stage of rice as examples:

- Zone A (Precision Regulation): Linked with sensor data, the system automatically triggers precision drip irrigation/watering to stabilize the micro-environment whenever anomalies in soil moisture or salinity (EC value) are detected.
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- Zone B (Traditional Flood Irrigation): Simulates conventional management by applying concentrated irrigation on a fixed 7-day schedule, completely independent of sensor data.

Verification Mechanism: During the experiment, the subterranean water and nutrient data (aligned automatically every 30 minutes in the cloud) will be cross-analyzed against the manual phenotypic data collected every 3 days (fruit diameter and cracking rate for persimmons; leaf area and growth vigor for rice). This cross-examination quantifies the actual efficacy of IoT-enabled precision regulation in improving the yield and quality of rice and persimmons. This partitioned experimental design enhances the scientific rigor of the findings and provides dependable data foundations for future large-scale adoption.

3. Data Collection Content and Gathering Strategies

To establish a comprehensive data ecosystem, this project collects data across three layers: environmental data, growth data, and outcome data.

Environmental Data

Environmental data is automatically captured by agricultural sensors, capturing continuous variables including soil moisture, soil temperature, soil pH, soil EC, air temperature, air humidity, light intensity, rainfall, and wind speed. Given the continuous nature of these variations, automated logging is utilized for long-term tracking.

In our 12-partition layout, each partition deploys one multi-channel sensor node tracking 6 core environmental metrics (soil temperature, soil moisture, EC value, air temperature, air humidity, and light intensity). Devices sample and upload data every 30 minutes, producing 48 records per device daily. Collectively, the entire experimental field generates 3,456 time-series environmental records per day. Across a single-season crop life cycle (estimated at 180 days), the system will stably accumulate approximately 622,000 fine-grained structured records. These continuous, high-frequency datasets will undergo structured storage and daily automated backups on the cloud IoT platform to serve as the foundation for subsequent analyses.

Crop Growth Data

Because certain critical growth indicators cannot be directly captured by IoT sensors, this project implements a "fixed-point random sampling" manual observation method to supplement phenotypic data accuracy. Technicians conduct field surveys every Monday morning strictly between 08:00 and 11:00 to eliminate physiological interferences caused by diurnal fluctuations in sunlight and transpiration.

Using a standard "five-point sampling method" within each partition, 5 fixed sample plants are tracked to continuously log plant height, leaf count, growth stage, flowering time of alkali persimmons, fruit count, and fruit size. These metrics are used to calculate the population's overall coverage rate and growth dynamics. During critical growth phases like fruit expansion, manual inspection frequency intensifies to once every 3 days. Additionally, a daily pest and disease screening is executed across the entire field at 16:00, recorded as binary calibration data (1 for occurrence, 0 for non-occurrence).

Outcome Data

During the harvest phase, final yield and quality metrics are recorded. Yield metrics include yield per plant, yield per *mu*, and total yield; quality metrics encompass sugar content (Brix), fruit size, visual appearance score, and product grading. Concurrently, commercial parameters such as selling price and revenue are documented, achieving full-process traceability from environmental inputs to economic benefits.

Data Quality Control Mechanism

To ensure the accuracy and reliability of analytical outputs, a data quality control mechanism will be established. Long-term operation of field sensors can be compromised by weather extremes, equipment aging, power fluctuations, and network transmission drops, leading to data gaps or anomalies. Therefore, systematic data quality assessments are mandatory prior to formal analysis. Short-term missing data will be imputed using adjacent-point data or historical averages, while values deviating severely from normal ranges will be identified and corrected against historical records. Meanwhile, a routine sensor inspection and calibration schedule will be instituted to maintain equipment stability. This rigorous data quality management guarantees database integrity, providing a dependable foundation for yield prediction, quality forecasting, and risk warnings.

4. Agricultural Big Data Analytics Scheme

Agricultural big data analytics stands as the core of this project, converting raw data into actionable management decisions.

Data Preprocessing and Descriptive Statistics

The initial stage focuses on data preprocessing. Raw data is cleaned and organized via missing value imputation, outlier identification, data standardization, and time-series alignment to establish a high-quality agricultural database. Following preprocessing, descriptive statistical analysis will be performed to decode the environmental variations across different periods and plots. This includes analyzing annual soil moisture trends, temperature variations, and the impact of rainfall on the soil profile. Trend graphs and distribution charts will be plotted to provide a preliminary understanding of data characteristics.

Factor Analysis and Predictive Modeling

Building on descriptive statistics, factor analysis will utilize correlation and regression techniques to evaluate the relationships between environmental variables and ultimate yield/quality. For instance, the analysis will quantify whether soil moisture correlates positively with final yield, how light duration governs product quality formation, and if soil salinity fluctuations affect crop growth velocity. Quantitative analysis will isolate dominant key variables and assess their impact magnitudes.Using environmental and growth data as input variables and final yield as the target variable, a yield prediction model will be developed. Trained on historical data, this model will enable early yield forecasting mid-season, allowing farmers to arrange marketing, storage, and logistics strategies well ahead of harvest.

Concurrently, a quality prediction model will be constructed to forecast final quality grades based on environmental conditions and growth metrics. Predicting parameters like sugar content and commercial grade based on temperature, humidity, and light exposure during the growth period allows farmers to fine-tune management practices early to maximize premium crop yields. Furthermore, an agricultural risk warning model will analyze historical anomalies to identify risk factors that trigger yield or quality degradation. The system will trigger automated alerts the moment real-time monitoring data crosses these pre-defined risk thresholds.

5. Agricultural Decision Support System Construction

The ultimate goal of data analytics is not merely generating research findings, but guiding actual on-farm production decisions. Thus, this project will construct an agricultural decision support platform designed to translate intricate data models into accessible management recommendations.The system automatically generates irrigation recommendations derived from real-time monitoring data. For example, when soil moisture falls below the optimal threshold, irrigation alerts are triggered; conversely, during heavy rainfall events, scheduled irrigation is suspended to eliminate water resource waste. If prolonged high-temperature and high-humidity conditions occur, the system issues reminders for pest and disease prevention, shifting farm management from reactive mitigation to proactive prevention.

The system also generates fertilization recommendations. By evaluating soil environmental dynamics to determine real-time nutrient demands, it identifies optimal fertilization windows, thereby boosting fertilizer use efficiency. This decision support system digitizes and intellectualizes farm management, driving down production costs while expanding operational efficiency.

6. Agricultural Data Visualization Platform Construction

To maximize the practical application of data insights, an agricultural data visualization platform will be developed. This platform will present the entire lifecycle of agricultural production through charts, dashboards, and spatial maps, enabling real-time status monitoring of the experimental plots.Managers can monitor soil moisture, temperature, rainfall, light intensity, and crop growth dynamics through desktop or mobile terminals. Should any monitored index exceed designated thresholds, the system immediately displays risk warnings through integrated graphics and text. For instance, persistent low soil moisture triggers a water-scarcity alert, while sustained hot, humid weather patterns trigger disease risk notifications. Visual dashboards present complex analytical models in an intuitive manner, allowing agricultural managers to accelerate decision-making efficiency.

7. Expected Outcomes

Upon completion, this project expects to deliver an integrated suite of outcomes, including an agricultural sensor monitoring platform, a comprehensive agricultural database, yield and quality prediction models, and an agricultural decision support system. Long-term data accumulation and continuous modeling will isolate the critical environmental factors dictating yield and quality, supplying solid scientific foundations for agricultural production.

From a practical perspective, the project is poised to optimize resource utilization efficiency, refine irrigation and fertilization management, minimize pest and disease losses, boost agricultural product quality and market competitiveness, and ultimately realize increased yields and income for farmers. From a research perspective, this work provides authentic case studies and datasets for agricultural big data analytics, smart agriculture, and digital farming, establishing a replicable blueprint for future digital transformations in agriculture.

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